

Personalized Hip Assistance to Improve Gait Performance in Individuals with Parkinson's Disease

Seonjeong Kim

Seoul National University

Seungjoo Noh

Seoul National University

Seo Jung Yun

Seoul National University Hospital

Jinwoo Hwang

Carnegie Mellon University

Yujin Kwon

UNC Chapel Hill & NC State University

Bokman Lim

Korea University of Technology and Education

Younbaek Lee

WIRobotics Inc

Han Gil Seo

Seoul National University Hospital

Keewon Kim

Seoul National University Hospital

Jinsoo Kim

jinskim@snu.ac.kr

Seoul National University

Research Article

Keywords: Parkinson's Disease, gait impairment, wearable robotics, hip assistance, personalized assistance, human-in-the-loop optimization, stride length, freezing of gait

Posted Date: August 21st, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-10612204/v1>

License: © ⓘ This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: Competing interest reported. This work was partially supported by WIRobotics through the provision of the wearable robots (WIM) and technical support. Han Gil Seo serves as an Associate Editor for the Journal of NeuroEngineering and Rehabilitation. The remaining authors declare that they have no competing interests.

Abstract

Background

Parkinson's disease (PD) is a progressive neurodegenerative disorder that commonly causes gait impairments, including reduced stride length, shuffling gait, festination, stooped posture, and freezing of gait (FoG). Conventional interventions can partially improve these gait abnormalities, but their overall effectiveness remains limited by side effects and inconsistent improvements in gait. To address these limitations, this study investigated whether personalized hip assistance delivered by a wearable robotic device could increase stride length and improve overall gait performance in individuals with PD.

Methods

On the assistance personalization day, human-in-the-loop optimization was used to identify personalized hip assistance profiles for 10 individuals with PD, with the objective of increasing stride length toward healthy control levels. Gait outcomes measured using inertial measurement units were compared between walking with and without assistance. On the biomechanical evaluation day, whole-body gait kinematics were assessed using a markerless motion capture system during single- and dual-task walking under three conditions: device not worn, device worn with assistance, and device worn without assistance.

Results

On the assistance personalization day, stride length increased significantly by $23.0 \pm 32.9\%$ with assistance compared with no assistance ($p < 0.05$). This improvement was accompanied by improved cadence and stride time variability, as well as increased hip range of motion and foot-to-floor angle. Moreover, FoG was substantially reduced in one participant. On the subsequent biomechanical evaluation day, the kinematic improvements extended beyond the assisted hip joint to the knee and ankle and were accompanied by reduced trunk flexion angle. Similar improvements in gait metrics were maintained during dual-task conditions.

Conclusions

These findings demonstrate that personalized hip assistance can effectively improve gait performance in individuals with PD while promoting whole-body biomechanical improvements beyond the assisted joint. Our results highlight the potential of personalized assistance to address the heterogeneous motor impairments associated with PD and may provide a generalizable framework for wearable robotic interventions across neurological populations with gait dysfunction.

Trial registration

This study was retrospectively registered in the Clinical Research Information Service (CRIS), Republic of Korea (registration number: KCT0012394).

Introduction

Parkinson's disease (PD) is one of the most common neurodegenerative disorders, affecting approximately 2–3% of individuals over the age of 65 [1, 2]. PD is characterized by the progressive loss of dopaminergic neurons in the substantia nigra, leading to dopamine deficiency in the striatum [3]. This deficiency results in dysfunction of the basal ganglia motor circuit and causes characteristic motor impairments. These motor impairments manifest as hypokinesia, bradykinesia, rigidity, and tremor, leading to abnormal gait patterns [4]. Individuals with PD exhibit various gait abnormalities, including reduced stride length, shuffling gait, festination, low foot clearance, and stooped posture [5, 6]. As the disease progresses, individuals with PD may also experience freezing of gait (FoG), a phenomenon in which forward progression temporarily ceases during walking despite the intention to walk [7]. Consequently, these gait impairments reduce mobility, increase the risk of falls [8], and limit independence in activities in daily living, ultimately affecting overall quality of life [9, 10]. These consequences highlight the need to address gait impairments in individuals with PD.

To address these gait impairments, conventional interventions for managing motor symptoms in PD can help improve gait abnormalities, but overall effectiveness remains limited by side effects and inconsistent improvement in gait. Among these, levodopa-based pharmacological treatment is widely used to manage motor symptoms in PD, as levodopa is converted into dopamine in the brain, replenishing dopamine levels and reducing motor symptoms [11]. With long-term use, however, patients may experience motor fluctuations and dyskinesia, which are common complications of levodopa therapy [12]. When the disease progresses or symptoms become refractory to pharmacological treatment, deep brain stimulation may be considered. It involves the implantation of electrodes into specific brain regions to modulate abnormal neural activity [13]. However, the implantation procedure is invasive and may be associated with adverse effects, including cognitive decline and psychiatric complications [14, 15]. Therefore, an alternative approach that can provide consistent and sustained improvements in gait is needed.

As an alternative approach, wearable robots have emerged as a promising method for gait assistance, but their effectiveness in individuals with PD remains to be fully established. Wearable robots provide direct assistance during walking through externally applied forces and have demonstrated benefits in improving gait performance across multiple studies [16–20]. In particular, such benefits have been reported in various clinical populations, including individuals with stroke, cerebral palsy, and spinal cord injury [21–24]. In the context of PD, a proof-of-concept study by Kim et al. [25] involving a single participant showed that robotic assistance could significantly reduce FoG episodes, along with improvements in other gait-related metrics, suggesting its potential to enhance gait performance. Given that this study involved a single participant, the generalizability of these findings remains limited, highlighting the need to evaluate robotic assistance in larger and more diverse populations to extend its clinical applicability.

Moreover, given the substantial inter-individual variability in gait patterns among individuals with PD, personalized assistance may help ensure consistent effectiveness across this population. Prior studies have shown that personalized optimization in robotic assistance further augments gait performance improvement. Studies in healthy individuals, who exhibit relatively low inter-individual variability in gait patterns, have demonstrated that individualized strategies yield greater benefits than fixed assistance approaches [26–28]. In individuals with PD, gait patterns vary considerably due to differences in disease severity and other clinical factors [29–31]. Therefore, personalized optimization may be particularly beneficial in the PD population.

To enable effective personalized optimization, stride length is a key gait parameter that reflects overall gait performance and can serve as a suitable target for optimization. Reduced stride length is a primary indicator of hypokinesia, one of the main motor symptoms of PD. Accordingly, individuals with PD show greater difficulty in regulating stride length compared to other gait parameters, characterized by reduced magnitude and increased variability [32–34]. Previous studies have shown that when stride length is temporarily improved using external cues, other gait parameters also show improvement, highlighting the importance of stride length in overall gait performance [33, 35]. A prior single-participant study by Kim et al. [25] further supports this, suggesting that the mitigation of FoG with robotic assistance may be attributed to its ability to prevent reductions in stride length and maintain it at an elevated level. Given the clinical significance of this parameter, optimizing robotic assistance based on stride length may provide a practical and clinically meaningful approach for improving gait performance in individuals with PD.

Whether the benefits of robotic assistance are maintained under increased cognitive demands remains an important question. Because gait automaticity is impaired in individuals with PD, gait control becomes increasingly attention-demanding when cognitive resources must be shared with a secondary task [36]. As a result, gait performance deteriorates during dual-task walking, accompanied by greater reductions in walking speed and stride length, along with increased variability [37, 38]. Importantly, dual-task walking can reveal gait impairments that are not apparent during single-task walking, providing a more sensitive assessment of functional gait deficits in PD [39]. Therefore, evaluating robotic assistance under dual-task conditions provides a more comprehensive assessment of its effectiveness than evaluation during controlled single-task walking alone.

In this study, we aim to evaluate the clinical effectiveness of personalized hip assistance using a wearable robot in individuals with PD by targeting increases in stride length toward levels observed in healthy individuals. We investigate whether personalized optimization leads to meaningful improvements in stride length and whether these improvements extend to other gait parameters and biomechanical outcomes, as improvements in stride length alone may not be sufficient to ensure overall gait quality. Based on this, we hypothesize that personalized optimization will lead to meaningful improvements in stride length and that such improvements will also be reflected in other gait parameters and biomechanical outcomes. Furthermore, the effects of robotic assistance are assessed under dual-

task conditions, with the hypothesis that these effects will be maintained even under increased task demands.

Methods

Study design and participants

The study was a prospective, single-center, exploratory, cross-sectional study approved by the Institutional Review Board of Seoul National University Hospital (IRB No. 2410-163-1582). This study was retrospectively registered with the Clinical Research Information Service (CRIS), Republic of Korea (KCT0012394). All procedures were conducted in accordance with the principles of Good Clinical Practice and the Declaration of Helsinki.

The inclusion criteria for participants with idiopathic PD were: (i) age ≥ 19 years, (ii) clinical diagnosis of idiopathic PD according to the UK Parkinson's disease Society Brain Bank Diagnostic Criteria, (iii) modified Hoehn and Yahr (mH&Y) stage 2.5 to 4, (iv) the ability to walk independently on level ground without assistive devices, and (v) the presence of gait abnormalities, such as shortened stride length, shuffling gait, or FoG. Exclusion criteria were: (i) a Korean version of the Mini-Mental State Examination (MMSE-K) score < 24 , (ii) neurological or orthopedic disorders other than PD that could affect gait (e.g., stroke, traumatic brain injury), and (iii) severe dyskinesia or on-off fluctuations.

Participants were recruited from an outpatient clinic at Seoul National University Hospital between February 2025 and May 2025, with the last follow-up date on June 27, 2025. All the participants provided written informed consent. Every study visit proceeded at the “on” state, which is the peak effect of PD medication.

Robotic Device

The wearable robot used in this study was WIM (WIRobotics Inc., Republic of Korea), designed to provide bilateral hip flexion and extension assistance during walking [40]. The device provides flexion torque to the swing leg and extension torque to the stance leg with the same magnitude. The assistance torque is governed by two control parameters that determine the magnitude and timing of assistance, referred to as gain (κ) and time delay (β), respectively. The device is controlled using an adaptive delayed output feedback control scheme. The current hip state $u(t_i)$ is determined from the current inter-thigh angle $\theta(t_i)$, defined as $u(t_i) = -2\sin(\theta(t_i)/2)$, where t_i refers to the current time. The filtered control state is given by $s(t_i) = 0.05u(t_i) + 0.95s(t_{i-1})$, where $s(t_i)$ represents the filtered version of $u(t_i)$. The assistive torque is generated based on the selected state timing as $\tau(t_i) = \kappa(0.8s(t_{i-\beta}) + 0.2s(t_{i-\beta-1}))$, where κ scales the torque magnitude, and β determines the temporal delay by reflecting β th past states relative to the current state. Given that the interval between consecutive indices (i and $i-1$) is 0.01s, this corresponds to an actual time delay of $\beta * 0.01$ s (Fig. 1A).

Experiment Protocol

The overall study scheme is as follows (Fig. 2B). A total of three experimental phases were conducted to evaluate the effects of personalized hip assistance. Participants meeting the inclusion and exclusion criteria were recruited during the Screening Day. Subsequently, a two-day Optimization phase was conducted, in which personalized assistive torque profiles were identified for each participant. Finally, during the Biomechanics Evaluation Day, the personalized hip assistance was applied to a subset of participants, and comprehensive gait assessments were performed.

● Screening Day

All participants received a detailed explanation of the study procedures, and written informed consent was obtained prior to participation. To facilitate adaptation to the robotic device, participants underwent a familiarization session. Participants completed a familiarization session until they reported no discomfort and could walk naturally without compensatory movements, even when the assistance magnitude and timing were adjusted.

● Optimization Day

The Optimization Day was conducted over two separate days. The experimental environment was a hospital corridor with a total distance of 128 m (Fig. 2A).

Prior to each day of the optimization process, participants underwent another robot familiarization session to adapt to the wearable device. During this session, they were instructed to walk naturally while following the assistive forces provided by the device in order to minimize initial resistance or discomfort. Specifically, the following verbal instructions were provided: “While walking with the robot, you may experience both assistance and resistance. Please walk in accordance with the assistance provided and avoid intentionally resisting any perceived interference. Instead, follow the forces generated by the robot and walk as naturally as possible.”

In this study, the gain (κ) and the time delay (β) were optimized within predefined parameter ranges of 5.5–30 and 0–40, respectively, which were determined to avoid interference with gait. To identify the optimal assistive profile for each individual, a human-in-the-loop optimization was implemented using the covariance matrix adaptation evolution strategy algorithm (Fig. 1B). The optimization aimed to minimize the difference between the participant’s current stride length and target stride length. The target stride length was determined based on reference values from healthy individuals matched for age, sex, and height [41]. It was introduced to prevent excessive increases in stride length, as unconstrained increases in stride length may adversely affect other gait parameters.

The optimization procedure was initially designed with 46 walking trials and was subsequently extended to 48 trials (six sets of eight), with each trial lasting 40 seconds. The detailed information regarding the

convergence process and criteria is provided in the Supplementary Methods. Convergence of the optimization process was defined as the standard deviation of the mean values across successive generations falling within a predefined resolution range, with step sizes of 1 for gain and 5 for time delay. The step sizes were established based on the minimum perceptible changes identified through empirical assessment by two members of the study team while wearing the device. However, the procedure was terminated at the corresponding generation if participant fatigue was reported or if physical limitations were identified based on clinical judgment. Additionally, in cases where the optimization converged to a local minimum, the process was restarted.

Following the completion of the optimization, a comparative gait validation was performed under two conditions: (1) ASSIST OFF (device worn but powered off) and (2) ASSIST ON (personalized optimal assistance applied), with the order of conditions randomized (Fig. 2A). To ensure participant safety and minimize the risk of falls during all gait tasks, participants wore a safety belt, and all procedures were supervised by licensed therapists.

● Biomechanics Evaluation Day

Based on the personalized assistance parameters obtained through the optimization process, biomechanical evaluation was conducted using a 2-minute walking test (2MWT). Participants walked along a track with a straight length of 10.8 m and a turning radius of 3.35 m, resulting in a total path length of approximately 39.4 m per traversal (Fig. 2C).

Two types of tasks were performed: single-task and dual-task. The dual-task trials were conducted after completion of the single-task trials. A total of three experimental conditions were evaluated: (1) ASSIST ON, (2) ASSIST OFF, and (3) NO EXO (device not worn). Each condition was tested under both single-task and dual-task conditions. Within each task, the NO EXO condition was always performed first and last, while the order of the remaining conditions was randomized (Fig. 2C).

To minimize potential adaptation effects associated with the device due to the time interval between the Optimization Day and the Biomechanics Evaluation Day, an additional robot familiarization session was conducted prior to the 2MWT.

In the dual-task conditions, participants simultaneously performed a motor task (2MWT) and a cognitive task (forward digit span). Each participant's maximum cognitive span was first determined in a seated resting condition. During the gait trials, participants performed the cognitive task at their individual maximum capacity by repeating digit sequences while walking, thereby inducing cognitive–motor interference. Given the potential risk of falls, participants at high risk of falling were secured with a safety belt and performed the 2MWT under the supervision of licensed therapists.

Data Collection

Baseline characteristics of the participants were collected as follows: age, sex, height, weight, disease duration, mH&Y stage, MDS-Unified Parkinson's Disease Rating Scale (MDS-UPDRS), New Freezing of Gait Questionnaire (NFOG-Q), and MMSE-K.

During the Optimization Day, four inertial measurement units (IMUs) (Xsens DOT, Movella Inc., USA) were attached—one to each lateral malleolus and one to each lateral thigh—to collect gait data.

On the Biomechanics Evaluation Day, IMUs were attached in the same configuration as the Optimization Day. Additionally, a markerless motion capture system consisting of ten cameras (Optitrack Prime Color, NaturalPoint, Inc., USA) was used to record whole-body gait kinematics during walking. Gait data were analyzed using markerless motion capture software (Theia, Theia Markerless Inc., Canada). After each condition of 2MWT, the remaining distance was measured using a distance measuring wheel.

Upon completion of all experimental sessions, participants completed a self-administered questionnaire assessing the usability of the wearable hip assistance robot. The questionnaire comprised seven items rated on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree), addressing overall satisfaction, the appropriateness of the device as an assistive aid for individuals with PD, ease of donning, physical comfort during use, perceived safety, willingness to continue using the device, and expected benefit for future gait function.

Outcome Measures

IMUs were used to compute spatiotemporal gait parameters during both the Optimization and Biomechanics Evaluation Days. Stride length was defined as the distance between successive midstance positions. In addition, walking speed, cadence, gait variability, hip range of motion, foot-to-floor angle, and toe clearance were computed. Detailed descriptions of the computation of each spatiotemporal gait parameter are provided in the Supplementary Methods.

During the Biomechanics Evaluation Day, motion capture data were additionally used to analyze joint angles and angular velocities of the hip, knee, and ankle. Furthermore, step length, step width, toe clearance, foot-to-floor angle, trailing limb angle, and average trunk flexion were computed.

Statistical Analysis

Paired t-tests were used to compare gait parameters between the ASSIST OFF and ASSIST ON conditions during the Optimization Day. For the biomechanical evaluation, statistical significance across the three conditions (NO EXO, ASSIST OFF, and ASSIST ON) was assessed using repeated-measures analysis of variance with a significance level set at $p < 0.05$. When significant effects were observed, post hoc pairwise comparisons were performed using paired t-tests.

Results

Baseline Characteristics

A total of 15 individuals were screened for eligibility, of whom 2 were excluded for not meeting the inclusion/exclusion criteria. The remaining 13 participants were enrolled in the study. One participant withdrew consent before the Optimization Day and was excluded, resulting in 12 participants who completed the study. On the Biomechanics Evaluation Day, data from 2 participants were lost owing to device malfunction; therefore, the final analysis included data from 10 participants (5 females). The mean age was 69.8 ± 6.6 years, and the mean disease duration was 108.8 ± 54.7 months. The mH&Y stage was 2.5 in five participants, 3 in four participants, and 4 in one participant. Detailed baseline characteristics of the participants are summarized in Table 1.

Table 1
Baseline characteristics of participants

ID	Age (year)	Sex	Height (m)	Mass (kg)	mH&Y stage	UPDRS- III	NFOG- Q	Target stride length (m)
1	77	M	1.56	53.0	4	54	23	1.31
2	55	F	1.56	52.9	3	30	0	1.29
3	69	M	1.74	85.8	2.5	21	0	1.45
4	67	M	1.76	65.5	3	47	7	1.47
5	71	F	1.55	62.2	3	42	13	1.24
6	75	F	1.53	48.7	3	41	26	1.22
7	64	F	1.55	42.7	2.5	17	0	1.26
8	72	M	1.71	73.4	2.5	18	0	1.42
9	72	F	1.46	41.5	2.5	37	0	1.17
10	76	M	1.70	68.6	2.5	21	0	1.41

mH&Y: modified Hoehn and Yahr; UPDRS-III: Unified Parkinson's Disease Rating Scale Part III; NFOG-Q: New Freezing of Gait Questionnaire. Target stride length was defined as the average value reported in the literature for healthy individuals matched by age, sex, and height [41].

For the Biomechanics Evaluation Day, data from five participants were excluded (four due to technical issues and one due to incomplete trial execution; $n = 5$). The number and type of trials varied across participants depending on individual condition and task feasibility.

Optimization Results

Following optimization, assistance parameters approached different values across participants (Fig. 3A). Six participants completed all 8 generations, whereas the remaining four did not complete the

full optimization protocol. One participant (P04) converged to a local minimum that required reinitialization, while three participants (P01, P06, and P09) requested early termination due to fatigue (Supplementary Fig. 1). The resulting assistance torque profiles also differed across participants. The duration of the maximum allowable torque (6 Nm) varied across participants. In participants who did not reach or reached the maximum torque in only a few strides (P04 and P05), distinct assistance profiles were observed due to differences in peak torque magnitude and timing (Fig. 3B, C).

Gait Performance under Single-task Conditions

Optimized assistance led to changes in multiple IMU-based gait metrics during single-task walking. Gait performance was evaluated in 10 participants during single-task walking on the Optimization Day, comparing the ASSIST OFF and ASSIST ON conditions over 40-s walking trials (Fig. 4). Stride length increased significantly by $23.0 \pm 32.9\%$ in the ASSIST ON condition compared to the ASSIST OFF condition ($p < 0.05$, Cohen's $d = 0.950$), with improvements observed in all participants. Cadence decreased significantly from 128 ± 19 to 116 ± 9 steps/min ($p < 0.05$, Cohen's $d = -0.880$), approaching values reported for healthy adults (107 steps/min), while overall walking speed remained unchanged. For the hip joint that received direct exoskeleton assistance, the range of motion increased significantly from $28.3 \pm 7.5^\circ$ to $34.4 \pm 7.2^\circ$ ($p < 0.05$, Cohen's $d = 1.001$). The foot-to-floor angle increased from $6.6 \pm 3.5^\circ$ to $12.2 \pm 4.6^\circ$ ($p < 0.001$, Cohen's $d = 1.621$), indicating a shift toward a heel-strike gait pattern. Stride time variability decreased from $4.1 \pm 2.7\%$ to $2.0 \pm 1.1\%$, suggesting a trend toward more stable gait rhythm.

In one participant with FoG, stride length across the walking trial was substantially increased under the ASSIST ON condition (Fig. 5A). No freezing episodes were observed under the ASSIST ON condition, whereas freezing occurred during 20% of the walking time under the ASSIST OFF condition (Fig. 5B; Supplementary Video 1).

Similar findings were observed during the Biomechanics Evaluation Day, where walking duration was extended from 40 seconds to 2 minutes and an additional NO EXO condition was included. When stride length was first evaluated in the five participants included in the Biomechanics Evaluation Day analysis under these conditions, it increased consistently in four participants in the ASSIST ON condition compared to the NO EXO and ASSIST OFF conditions, while the remaining participant showed no increase, and the stride length variability decreased in the same four participants (Supplementary Fig. 2). As the optimization primarily aimed to improve stride length, further analyses were performed in these four participants to examine changes in additional gait metrics.

Among these participants, cadence shifted toward values reported for healthy adults, and hip range of motion increased under ASSIST ON condition compared to the NO EXO and ASSIST OFF conditions. Stride time variability decreased, and stride length variability also shifted toward values reported for healthy adults (Supplementary Fig. 3; Supplementary Video 2). These results were obtained over longer walking durations (2 min compared to 40 s), further supporting the robustness of the observed

improvements. The inclusion of the NO EXO condition enabled comparison with a reference condition without the wearable device.

Biomechanical Analysis under Single-task Conditions

Motion capture-based biomechanical analysis further characterized changes in lower-limb kinematics and trunk posture with optimized assistance. The analysis was performed in two participants with valid motion capture data from the Biomechanics Evaluation Day who exhibited improvement in stride length with optimized assistance. At the hip joint receiving direct robotic assistance, range of motion increased, consistent with IMU-based findings. Peak extension velocity during the loading response and peak flexion velocity during the pre-swing phase also increased (Fig. 6A). In addition, at the knee, range of motion increased along with peak flexion velocity during the swing phase (Fig. 6B). At the ankle, range of motion increased along with peak plantarflexion velocity during the push-off phase (Fig. 6C).

Additional metrics are shown in Fig. 6D. Step length and toe clearance increased with optimized assistance. Consistent with the IMU-based results, the foot-to-floor angle increased under the ASSIST ON condition. Average trunk flexion decreased, indicating a shift toward a more upright posture. Step width and trailing limb angle did not show consistent trends across the two participants. These measures reflect gait stability and the capacity to generate propulsion during push-off, respectively.

Gait Performance under Dual-task Conditions

Optimized assistance effects were also observed under dual-task conditions. IMU-based gait parameters were analyzed in three participants who completed the dual-task condition among the four participants included in the single-task analysis. Compared to single-task walking, walking speed generally decreased and stride length was reduced under dual-task conditions (Fig. 7A), indicating higher task demands.

Across the NO EXO, ASSIST OFF, and ASSIST ON conditions, stride length was greater in the ASSIST ON condition than in the other two conditions. Cadence did not show consistent changes across participants. Other gait parameters showed similar trends across the three conditions, consistent with the single-task results. Hip range of motion and foot-to-floor angle increased with assistance, while stride length variability and stride time variability decreased. Toe clearance increased in two of the three participants (P03 and P04). Digit span test accuracy decreased under the ASSIST ON condition in all participants (Fig. 7B).

Usability Assessment

All 10 participants completed the usability questionnaire. Ratings were favorable across all seven items (overall mean, 5.4 ± 1.0). Overall satisfaction with the device was rated 5.6 ± 0.8 , and participants rated the device as appropriate for assisting gait in individuals with PD (5.5 ± 0.5). Physical comfort and perceived safety during use were rated 5.5 ± 1.3 and 5.2 ± 1.0 , respectively, while ease of donning

received the lowest rating among the seven items (4.5 ± 1.4). Willingness to continue using the device was rated 5.5 ± 0.8 , and expected benefit for future gait function received the highest rating (5.9 ± 0.7).

Discussion

This study demonstrates that personalized hip assistance delivered by a wearable robot can improve gait performance in individuals with PD by increasing stride length toward levels observed in healthy individuals. Importantly, the benefits extended beyond the optimization target of stride length, resulting in broader improvements in gait performance. Moreover, the improvements were maintained during dual-task walking, demonstrating that the benefits of robotic assistance persist under increased task demands and providing a more rigorous evaluation of its effectiveness. Consistent with these objective gait improvements, participants also reported favorable usability ratings, including high overall satisfaction and expected benefit for future gait function, supporting the acceptability of the device among individuals with PD.

The optimization process resulted in participant-specific assistance strategies, with stride length increased in all participants under optimized assistance. From a biomechanical perspective, greater stride length is associated with increased joint moments [45], and previous studies have reported that higher assistance magnitude leads to greater range of motion at the targeted joint [46, 47]. However, in our study, larger assistance magnitude did not necessarily produce greater stride length. This finding indicates that the effectiveness of robotic gait assistance depends not only on assistance magnitude, but also on multiple factors, including temporal characteristics and overall assistance profile. This multi-factor dependency provides a strong rationale for personalized optimization over simply increasing assistance magnitude.

To examine whether the optimized assistance strategies translated into broader improvements in gait performance, we evaluated additional gait metrics beyond stride length. Several gait metrics improved under optimized assistance, suggesting that optimization focused on stride length may extend to broader improvements in overall gait patterns. During single-task walking, the increase in stride length was accompanied by a significant reduction in cadence, approaching values observed in healthy individuals. As increased cadence in PD is often considered a compensatory response to reduced stride length [32, 33], this finding suggests a shift toward a more normalized gait pattern. Our findings are consistent with previous studies using visual cues to increase stride length in individuals with PD, which have also reported improvements in gait parameters beyond stride length. Morris et al. and Lewis et al. [33, 35] reported a shift in cadence toward values observed in healthy individuals when stride length was increased using visual cues. In addition, Lewis et al. [30] reported findings similar to ours, including improvements in hip kinematics and heel rocker function, as well as a reduction in stride length variability. However, previous studies reported increases in walking speed, whereas no significant change was observed during robotic assistance in our study.

Robotic assistance also prevented FoG in the present study. One of the 10 participants exhibited FoG during the experimental sessions, and no FoG episodes occurred in that participant while assistance was applied. This finding is consistent with a proof-of-concept study by Kim et al. [25]. One possible explanation relates to the dual-requirement hypothesis, which attributes FoG to the coexistence of a reduced step length and the sequence effect [48]. Assistance synchronized with the gait cycle and optimized toward a target stride length may have acted on both components. While the prior study applied hip flexion assistance alone, our study provided bilateral assistance for both hip flexion and extension, raising the possibility that successful FoG prevention may not depend on a specific joint or assistance strategy. However, both observations derive from single individuals, and given the heterogeneity of PD symptoms, the optimal joint or assistance strategy for preventing FoG may vary across individuals.

We further analyzed whole-body kinematics to examine how these improvements manifested in overall gait patterns and observed positive changes across multiple joints. Notably, kinematic improvements at non-targeted joints may be attributed to biomechanical coupling among lower-limb joints during gait [49, 50], and similar findings have been consistently reported in previous wearable robot studies [51–53].

To assess whether the effects of optimized assistance were maintained under more challenging conditions, gait performance was evaluated during dual-task walking. Improvements in gait metrics observed during single-task walking were largely preserved during dual-task walking. However, cognitive task performance declined with robotic assistance. This suggests that robotic assistance may have shifted attentional resources toward gait, leading participants to adopt a “posture-first” strategy, in contrast to the “posture-second” strategy typically observed in individuals with PD [54, 55].

Individual variability in response to optimized assistance

The extent of gait improvement varied across participants, suggesting that the effectiveness of robotic assistance differs across individuals. Therefore, identifying potential responders in advance may be important. However, no significant relationship was found between stride length improvement and clinical measures (i.e., UPDRS-III) or baseline gait performance (Supplementary Fig. 4). Meanwhile, the absence of FoG under robotic assistance suggests that this approach may be particularly beneficial for individuals with FoG symptoms. Since intervention effects may vary depending on individual patient characteristics [56–58], further studies examining multi-factorial effects may provide additional insight.

Challenges in finding optimal assistance

Identifying optimal assistance under strict convergence criteria is challenging in individuals with PD, as prolonged walking required for iterative optimization is often not feasible due to their limited walking endurance. We designed the optimization protocol to promote convergence within limited walking time by incorporating both the mean of the current sampling distribution and the best-performing parameters from the previous generation (Supplementary Methods). Nevertheless, practical constraints remain due to the limited walking endurance of individuals with PD compared with healthy controls. For example, it

is difficult to conduct additional evaluations to ensure full convergence, particularly when the solution may be suboptimal or trapped in a local minimum. Moreover, some control parameters may inherently require a large number of iterations to reach convergence [59]. These constraints underscore the need for more efficient human-in-the-loop optimization approaches that enable convergence with fewer evaluations.

Adaptive optimization may be necessary to achieve effective assistance in the presence of day-to-day variability and the effects of medication cycles. Gait characteristics in individuals with PD exhibit substantial day-to-day variability. Dopaminergic medication can substantially influence gait performance, and intrinsic motor fluctuations associated with PD can cause variable performance even under consistent medication conditions [60, 61]. This day-to-day variability was also observed in our study. Specifically, baseline stride length without robotic assistance varied across experimental days. Such variability may lead to mismatches between the participant's state during optimization and that during subsequent real-world use, potentially limiting the effectiveness of the optimized parameters. In our study, when the robotic assistance using identical optimized control parameters was applied on a different day, responses varied across participants (Supplementary Fig. 5), supporting this concern. These findings suggest that static optimization approaches may have inherent limitations in the presence of day-to-day variability, and adaptive or real-time optimization strategies [27] may be particularly crucial for individuals with PD, as they can track changes in the user's state and adjust assistance accordingly.

Contribution of mechanical assistance and cueing effects

The mechanisms underlying the effects of robotic gait assistance should be further examined to enhance and generalize the effectiveness of robotic gait assistance. Although the primary function of robotic assistance is to provide mechanical support, physical interaction between the device and the user may also act as a form of rhythmic tactile cue. Mechanical assistance increases movement amplitude through joint torque augmentation, which likely contributes to the observed increase in stride length. Beyond direct mechanical assistance, tactile cues may act as attentional signals, enhancing task engagement and thereby contributing to improvements in gait performance. Previous studies have shown that various forms of cueing (e.g., verbal, attentional, and visual cues) can enhance gait performance in individuals with PD [33, 35, 62]. Morris et al. [33] further proposed that these effects are attributable to cues directing attention to the walking task, rather than to the specific characteristics of the cues. In our dual-task results, assistance improved gait performance while cognitive task performance declined, suggesting that attentional resources were primarily allocated to walking, possibly reflecting a cueing effect. Meanwhile, Kim et al. [25] reported that cueing-level robotic assistance without mechanical support was insufficient to prevent FoG, suggesting that the contribution of cueing effects in robotic assistance requires further investigation.

Limitations

This study has several limitations. First, a commercial device was used, which limited independent control of the assistance torque profile. As the torque profile was coupled to the inter-thigh angle, it could vary even under identical control parameters, depending on individual gait patterns. Future systems that allow the torque profile to be independently specified from control parameters may enable more rigorous experimental control. Second, medication status was not strictly controlled. Although participants were tested in an “on” state, potential fluctuations or decay in medication effects may have influenced gait performance. Third, the personalized torque profile was fixed after optimization. Although improvements were observed on both days, the effects were less pronounced on the Biomechanical Evaluation Day than on the Optimization Day, suggesting that day-to-day variability in PD may influence the effectiveness of the optimized-but-fixed assistance. Future studies may explore adaptive control strategies to account for day-to-day variability in PD. Finally, although stride length increased with assistance, not all participants approached the target stride length. This variability further highlights the potential influence of individual characteristics on the effectiveness of robotic assistance and suggests that additional strategies may be required to further enhance outcomes. For example, combining robotic assistance with rehabilitation or providing longer-term exposure to allow sufficient adaptation may further improve effectiveness [63].

Conclusion

Providing personalized hip assistance targeting stride length improvement leads to enhanced overall gait performance in individuals with PD. Beyond the primary target, these improvements extend to multiple spatiotemporal and kinematic measures and are maintained under increased task demands, such as dual-task walking. These findings highlight the potential of personalized optimization to induce broader benefits in gait performance beyond a single targeted gait metric. Future work should focus on identifying individuals who are more likely to benefit from such interventions in advance, as well as developing adaptive optimization strategies that account for the day-to-day variability and heterogeneity of gait in individuals with PD, enabling more effective personalized assistance.

Abbreviations

2MWT

2-Minute Walk Test

FoG

Freezing of Gait

IMUs

Inertial Measurement Units

mH&Y

modified Hoehn and Yahr

MDS-UPDRS

MDS-Unified Parkinson's Disease Rating Scale

NFOG-Q

New Freezing of Gait Questionnaire
PD
Parkinson's Disease
ZUPT
Zero-Velocity Update

Declarations

Ethics approval and consent to participate

All experiments were approved by the Institutional Review Board (IRB) of Seoul National University Hospital (IRB No. 2410-163-1582), and informed consent was obtained from all participants.

Consent for publication

Informed consent for publication was obtained from all participants.

Competing Interests

This work was partially supported by WIRobotics through the provision of the wearable robots (WIM) and technical support. Han Gil Seo serves as an Associate Editor for the Journal of NeuroEngineering and Rehabilitation. The remaining authors declare that they have no competing interests.

Funding

This work was supported by WIRobotics Inc., the Creative Pioneering Researchers Program through Seoul National University, and the Interdisciplinary Research Initiatives Program of the College of Engineering and the College of Medicine, Seoul National University (2025–2026). This work was partly supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korean government (MSIT) [RS-2021-II211343, Artificial Intelligence Graduate School Program (Seoul National Univ.)] and [RS-2025-25442338, AI Star Fellowship Support Program (Seoul National Univ.)].

Author Contribution

S.K. and J.H. implemented and developed algorithms for the experiments. S.K., S.J.Y., J.H., H.G.S., K.K., and J.K. designed the experiments. S.K., S.N., S.J.Y., J.H., and Y.K. conducted the human-subject experiments. S.K. and S.N. analyzed the data, drafted and edited the manuscript. S.K. and S.N. prepared figures. B.L. and Y.L. provided technical support. H.G.S., K.K., and J.K. conceived and managed the project. All authors read and approved the final manuscript.

Acknowledgement

The authors are most grateful to all the patients for their commitment, patience, and dedication throughout the study. The authors also thank Jeong Eun Shin (Seoul National University Hospital, Seoul, Republic of Korea) for her assistance with data collection and Minjong Kim (Seoul National University, Seoul, Republic of Korea) for his assistance in validating the IMU-based algorithm against the motion capture system.

Data Availability

All data generated or analyzed during this study are included in this article.

References

1. Twelves D, Perkins KSM, Counsell C. Systematic review of incidence studies of Parkinson's disease. *Mov Disord*. 2003;18:19–31. <https://doi.org/10.1002/mds.10305>.
2. Savica R, Grossardt BR, Bower JH, Ahlskog JE, Rocca WA. Incidence and Pathology of Synucleinopathies and Tauopathies Related to Parkinsonism. *JAMA Neurol*. 2013;70:859–66. <https://doi.org/10.1001/jamaneurol.2013.114>.
3. Connolly BS, Lang AE. Pharmacological Treatment of Parkinson Disease: A Review. *JAMA*. 2014;311:1670. <https://doi.org/10.1001/jama.2014.3654>.
4. Bloem BR, Okun MS, Klein C. Parkinson's disease. *Lancet Elsevier*. 2021;397:2284–303. [https://doi.org/10.1016/S0140-6736\(21\)00218-X](https://doi.org/10.1016/S0140-6736(21)00218-X).
5. Ebersbach G, Moreau C, Gandor F, Defebvre L, Devos D. Clinical syndromes: Parkinsonian gait. *Mov Disord*. 2013;28:1552–9. <https://doi.org/10.1002/mds.25675>.
6. Morris ME, Huxham F, McGinley J, Dodd K, Iansek R. The biomechanics and motor control of gait in Parkinson disease. *Clin Biomech*. 2001;16:459–70. [https://doi.org/10.1016/S0268-0033\(01\)00035-3](https://doi.org/10.1016/S0268-0033(01)00035-3).
7. Nutt JG, Bloem BR, Giladi N, Hallett M, Horak FB, Nieuwboer A. Freezing of gait: moving forward on a mysterious clinical phenomenon. *Lancet Neurol*. 2011;10:734–44. [https://doi.org/10.1016/S1474-4422\(11\)70143-0](https://doi.org/10.1016/S1474-4422(11)70143-0).
8. Allen NE, Schwarzel AK, Canning CG. Recurrent Falls in Parkinson's Disease: A Systematic Review. *Park Dis*. 2013;2013:906274. <https://doi.org/10.1155/2013/906274>.
9. Schrag A, Jahanshahi M, Quinn N. What contributes to quality of life in patients with Parkinson's disease? *J Neurol Neurosurg Psychiatry*. BMJ Publishing Group Ltd. 2000;69:308–12. <https://doi.org/10.1136/jnnp.69.3.308>.
10. Wicks P, Stamford J, Grootenhuys MA, Haverman L, Ahmed S. Innovations in e-health. *Qual Life Res*. 2014;23:195–203. <https://doi.org/10.1007/s11136-013-0458-x>.

11. LeWitt PA, Fahn S. Levodopa therapy for Parkinson disease. *Neurol Wolters Kluwer*. 2016;86:S3–12. <https://doi.org/10.1212/WNL.0000000000002509>.
12. Jankovic J. Motor fluctuations and dyskinesias in Parkinson's disease: Clinical manifestations. *Mov Disord*. 2005;20:S11–6. <https://doi.org/10.1002/mds.20458>.
13. Benabid AL. Deep brain stimulation for Parkinson's disease. *Curr Opin Neurobiol*. 2003;13:696–706. <https://doi.org/10.1016/j.conb.2003.11.001>.
14. Limousin P, Foltynie T. Long-term outcomes of deep brain stimulation in Parkinson disease. *Nat Rev Neurol Nat Publishing Group*. 2019;15:234–42. <https://doi.org/10.1038/s41582-019-0145-9>.
15. Massano J, Garrett C. Deep Brain Stimulation and Cognitive Decline in Parkinson's Disease: A Clinical Review. *Front Neurol [Internet]*. Frontiers; 2012 [cited 2026 Apr 30];3. <https://doi.org/10.3389/fneur.2012.00066>
16. Collins SH, Wiggin MB, Sawicki GS. Reducing the energy cost of human walking using an unpowered exoskeleton. *Nature*. 2015;522:212–5. <https://doi.org/10.1038/nature14288>.
17. Kim J, Lee G, Heimgartner R, Arumukhom Revi D, Karavas N, Nathanson D, et al. Reducing the metabolic rate of walking and running with a versatile, portable exosuit. *Science*. 2019;365:668–72. <https://doi.org/10.1126/science.aav7536>.
18. Lee G, Kim J, Panizzolo FA, Zhou YM, Baker LM, Galiana I, et al. Reducing the metabolic cost of running with a tethered soft exosuit. *Sci Robot*. 2017;2:eaan6708. <https://doi.org/10.1126/scirobotics.aan6708>.
19. Molinaro DD, Scherpereel KL, Schonhaut EB, Evangelopoulos G, Shepherd MK, Young AJ. Task-agnostic exoskeleton control via biological joint moment estimation. *Nature*. 2024;635:337–44. <https://doi.org/10.1038/s41586-024-08157-7>.
20. Sawicki GS, Beck ON, Kang I, Young AJ. The exoskeleton expansion: improving walking and running economy. *J Neuroeng Rehabil*. 2020;17:25. <https://doi.org/10.1186/s12984-020-00663-9>.
21. Awad LN, Bae J, O'Donnell K, De Rossi SMM, Hendron K, Sloat LH, et al. A soft robotic exosuit improves walking in patients after stroke. *Sci Transl Med Am Association Advancement Sci*. 2017;9:eaai9084. <https://doi.org/10.1126/scitranslmed.aai9084>.
22. Hunt M, Everaert L, Brown M, Muraru L, Hatzidimitriadou E, Desloovere K. Effectiveness of robotic exoskeletons for improving gait in children with cerebral palsy: A systematic review. *Gait Posture*. 2022;98:343–54. <https://doi.org/10.1016/j.gaitpost.2022.09.082>.
23. Lerner ZF, Damiano DL, Bulea TC. A lower-extremity exoskeleton improves knee extension in children with crouch gait from cerebral palsy. *Sci Transl Med Am Association Advancement Sci*. 2017;9:eaam9145. <https://doi.org/10.1126/scitranslmed.aam9145>.
24. Miller LE, Zimmermann AK, Herbert WG. Clinical effectiveness and safety of powered exoskeleton-assisted walking in patients with spinal cord injury: systematic review with meta-analysis. *Med Devices Evid Res Dove Med Press*. 2016;9:455–66. <https://doi.org/10.2147/MDER.S103102>.
25. Kim J, Porciuncula F, Yang HD, Wendel N, Baker T, Chin A, et al. Soft robotic apparel to avert freezing of gait in Parkinson's disease. *Nat Med*. 2024;30:177–85. <https://doi.org/10.1038/s41591-023->

26. Ding Y, Kim M, Kuindersma S, Walsh CJ. Human-in-the-loop optimization of hip assistance with a soft exosuit during walking. *Sci Robot Am Association Advancement Sci*. 2018;3:eaar5438. <https://doi.org/10.1126/scirobotics.aar5438>.
27. Slade P, Kochenderfer MJ, Delp SL, Collins SH. Personalizing exoskeleton assistance while walking in the real world. *Nature*. 2022;610:277–82. <https://doi.org/10.1038/s41586-022-05191-1>.
28. Zhang J, Fiers P, Witte KA, Jackson RW, Poggensee KL, Atkeson CG, et al. Human-in-the-loop optimization of exoskeleton assistance during walking. *Science*. 2017;356:1280–4. <https://doi.org/10.1126/science.aal5054>.
29. Galna B, Lord S, Burn DJ, Rochester L. Progression of gait dysfunction in incident Parkinson's disease: Impact of medication and phenotype. *Mov Disord*. 2015;30:359–67. <https://doi.org/10.1002/mds.26110>.
30. Lewis SJG. Heterogeneity of Parkinson's disease in the early clinical stages using a data driven approach. *J Neurol Neurosurg Psychiatry*. 2005;76:343–8. <https://doi.org/10.1136/jnnp.2003.033530>.
31. Wolters ECh. Variability in the clinical expression of Parkinson's disease. *J Neurol Sci*. 2008;266:197–203. <https://doi.org/10.1016/j.jns.2007.08.016>.
32. Morris ME, Iansek R, Matyas TA, Summers JJ. The pathogenesis of gait hypokinesia in Parkinson's disease. *Brain*. 1994;117:1169–81. <https://doi.org/10.1093/brain/117.5.1169>.
33. Morris ME, Iansek R, Matyas TA, Summers JJ. Stride length regulation in Parkinson's disease: Normalization strategies and underlying mechanisms. *Brain*. 1996;119:551–68. <https://doi.org/10.1093/brain/119.2.551>.
34. Blin O, Ferrandez AM, Serratrice G. Quantitative analysis of gait in Parkinson patients: increased variability of stride length. *J Neurol Sci*. 1990;98:91–7. [https://doi.org/10.1016/0022-510X\(90\)90184-O](https://doi.org/10.1016/0022-510X(90)90184-O).
35. Lewis GN, Byblow WD, Walt SE. Stride length regulation in Parkinson's disease: the use of extrinsic, visual cues. *Brain*. 2000;123:2077–90. <https://doi.org/10.1093/brain/123.10.2077>.
36. Yogev G, Giladi N, Peretz C, Springer S, Simon ES, Hausdorff JM. Dual tasking, gait rhythmicity, and Parkinson's disease: Which aspects of gait are attention demanding? *Eur J Neurosci*. 2005;22:1248–56. <https://doi.org/10.1111/j.1460-9568.2005.04298.x>.
37. Kelly VE, Eusterbrock AJ, Shumway-Cook A. A Review of Dual-Task Walking Deficits in People with Parkinson's Disease: Motor and Cognitive Contributions, Mechanisms, and Clinical Implications. *Park Dis*. 2012;2012:918719. <https://doi.org/10.1155/2012/918719>.
38. Raffegeau TE, Krehbiel LM, Kang N, Thijs FJ, Altmann LJP, Cauraugh JH et al. A Meta-Analysis: Parkinson's Disease and Dual-Task Walking. 2021.
39. Wild LB, Lima DB, Balardin JB, Rizzi L, Giacobbo BL, Oliveira HB, et al. Characterization of cognitive and motor performance during dual-tasking in healthy older adults and patients with Parkinson's disease. *J Neurol*. 2013;260:580–9. <https://doi.org/10.1007/s00415-012-6683-3>.

40. Lim B, Choi B, Roh C, Lee J, Kim Y-J, Lee Y. Ultra-lightweight robotic hip exoskeleton with anti-phase torque symmetry for enhanced walking efficiency. *Sci Rep.* 2025;15:10850. <https://doi.org/10.1038/s41598-025-95599-2>.
41. Samson MM, Crowe A, De Vreede PL, Dessens JAG, Duursma SA, Verhaar HJJ. Differences in gait parameters at a preferred walking speed in healthy subjects due to age, height and body weight. *Aging Clin Exp Res.* 2001;13:16–21. <https://doi.org/10.1007/BF03351489>.
42. Vette AH, Watt JM, Lewicke J, Watkins B, Burkholder LM, Andersen J, et al. The utility of normative foot floor angle data in assessing toe-walking. *Foot.* 2018;37:65–70. <https://doi.org/10.1016/j.foot.2018.07.003>.
43. Hollman JH, McDade EM, Petersen RC. Normative spatiotemporal gait parameters in older adults. *Gait Posture.* 2011;34:111–8. <https://doi.org/10.1016/j.gaitpost.2011.03.024>.
44. Begg R, Best R, Dell'Oro L, Taylor S. Minimum foot clearance during walking: Strategies for the minimisation of trip-related falls. *Gait Posture.* 2007;25:191–8. <https://doi.org/10.1016/j.gaitpost.2006.03.008>.
45. Ardestani MM, Ferrigno C, Moazen M, Wimmer MA. From normal to fast walking: Impact of cadence and stride length on lower extremity joint moments. *Gait Posture.* 2016;46:118–25. <https://doi.org/10.1016/j.gaitpost.2016.02.005>.
46. Kim J, Quinlivan BT, Deprey L-A, Arumukhom Revi D, Eckert-Erdheim A, Murphy P, et al. Reducing the energy cost of walking with low assistance levels through optimized hip flexion assistance from a soft exosuit. *Sci Rep.* 2022;12:11004. <https://doi.org/10.1038/s41598-022-14784-9>.
47. Kang I, Hsu H, Young A. The Effect of Hip Assistance Levels on Human Energetic Cost Using Robotic Hip Exoskeletons. *IEEE Robot Autom Lett.* 2019;4:430–7. <https://doi.org/10.1109/LRA.2019.2890896>.
48. Chee R, Murphy A, Danoudis M, Georgiou-Karistianis N, Iansek R. Gait freezing in Parkinson's disease and the stride length sequence effect interaction. *Brain.* 2009;132:2151–60. <https://doi.org/10.1093/brain/awp053>.
49. Krasovsky T, Levin MF, Review, Inc STM. 2010;24:213–24. <https://doi.org/10.1177/1545968309348509>.
50. Winter DA. Foot Trajectory in Human Gait: A Precise and Multifactorial Motor Control Task. *Phys Ther.* 1992;72:45–53. <https://doi.org/10.1093/ptj/72.1.45>.
51. Franks PW, Bryan GM, Martin RM, Reyes R, Lakmazaheri AC, Collins SH. Comparing optimized exoskeleton assistance of the hip, knee, and ankle in single and multi-joint configurations. *Wearable Technol.* 2021;2:e16. <https://doi.org/10.1017/wtc.2021.14>.
52. Lerner ZF, Damiano DL, Bulea TC. The Effects of Exoskeleton Assisted Knee Extension on Lower-Extremity Gait Kinematics, Kinetics, and Muscle Activity in Children with Cerebral Palsy. *Sci Rep Nat Publishing Group.* 2017;7:13512. <https://doi.org/10.1038/s41598-017-13554-2>.
53. Lambranzi C, Oberti G, Natali CD, Caldwell DG, Galli M, De Momi E, et al. Impact of a Lower Limb Exosuit Anchor Points on Energetics and Biomechanics. *IEEE Trans Biomed Eng.* 2026;73:779–92.

<https://doi.org/10.1109/TBME.2025.3593040>.

54. Riedel N. An outdoor dual-task study on cognitive-motor interference during exoskeleton-assisted walking.
55. Bloem BR, Grimbergen YAM, Van Dijk JG, Munneke M. The posture second strategy: A review of wrong priorities in Parkinson's disease. *J Neurol Sci.* 2006;248:196–204.
<https://doi.org/10.1016/j.jns.2006.05.010>.
56. Kegelmeyer DA, Minarsch R, Kostyk SK, Kline D, Smith R, Kloos AD. Use of a Robotic Walking Device for Home and Community Mobility in Parkinson Disease: A Randomized Controlled Trial. *J Neurol Phys Ther.* 2024;48:102–11. <https://doi.org/10.1097/NPT.0000000000000467>.
57. Albers C, Mirelman A, Avanzino L, Bloem BR, Botta A, van der Crujisen J, et al. Understanding Cueing Strategies for Gait Impairments in Parkinson's Disease: Protocol of the Multicenter UNITE-PD Study. *Eur J Neurosci.* 2026;63:e70382. <https://doi.org/10.1111/ejn.70382>.
58. Li X, Wang S, Wang K, Wang W, Tuo H, Long Y, et al. Personalized auditory cues improve gait in patients with early Parkinson's disease. *Front Neurol [Internet] Front.* 2025.
<https://doi.org/10.3389/fneur.2025.1561880>. [cited 2026 Apr 30];16.
59. Poggensee KL, Collins SH. How adaptation, training, and customization contribute to benefits from exoskeleton assistance. *Sci Robot.* 2021.
<https://doi.org/https://doi.org/10.1126/scirobotics.abf1078>. 6.
60. Holleran CL, Bland MD, Reisman DS, Ellis TD, Earhart GM, Lang CE. Day-to-Day Variability of Walking Performance Measures in Individuals Poststroke and Individuals With Parkinson Disease. *J Neurol Phys Ther.* 2020;44:241–7. <https://doi.org/10.1097/NPT.0000000000000327>.
61. O'Sullivan JD, Said CM, Dillon LC, Hoffman M, Hughes AJ. Gait analysis in patients with Parkinson's disease and motor fluctuations: Influence of levodopa and comparison with other measures of motor function. *Mov Disord.* 1998;13:900–6. <https://doi.org/10.1002/mds.870130607>.
62. Behrman AL, Teitelbaum P, Cauraugh JH. Verbal instructional sets to normalise the temporal and spatial gait variables in Parkinson's disease. *J Neurol Neurosurg Psychiatry.* 1998;65:580–2.
<https://doi.org/10.1136/jnnp.65.4.580>.
63. Otlet V, Vandamme C, Warlop T, Crevecoeur F, Ronsse R. Effects of overground gait training assisted by a wearable exoskeleton in patients with Parkinson's disease. *J Neuroeng Rehabil.* 2023;20:156.
<https://doi.org/10.1186/s12984-023-01280-y>.

Figures

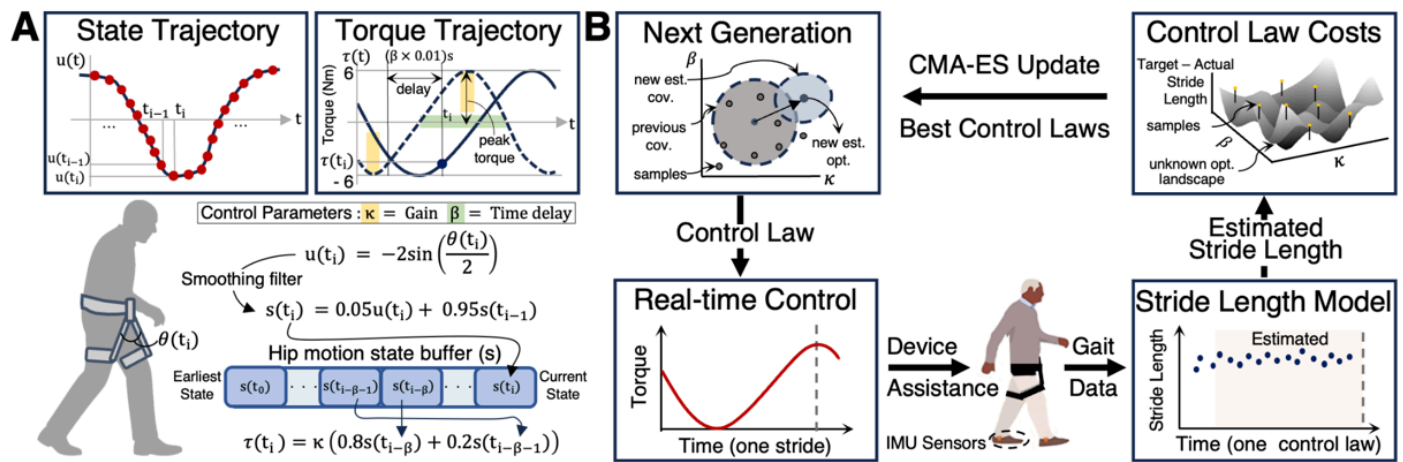


Fig.1 Operating principles of the WIM device and human-in-the-loop optimization framework. (A) Schematic of how the two optimization parameters—gain (κ) and time delay (β)—determine the torque profile in the robotic device. The yellow and green shaded regions indicate the allowable bounds of the gain (κ) and time delay (β) parameters, respectively. The optimization was performed within these bounds. The torque was saturated at 6 Nm, with values exceeding this limit clipped. (B) Workflow of the human-in-the-loop optimization process using covariance matrix adaptation evolution strategy (CMA-ES) algorithm.

Figure 1

See image above for figure legend.

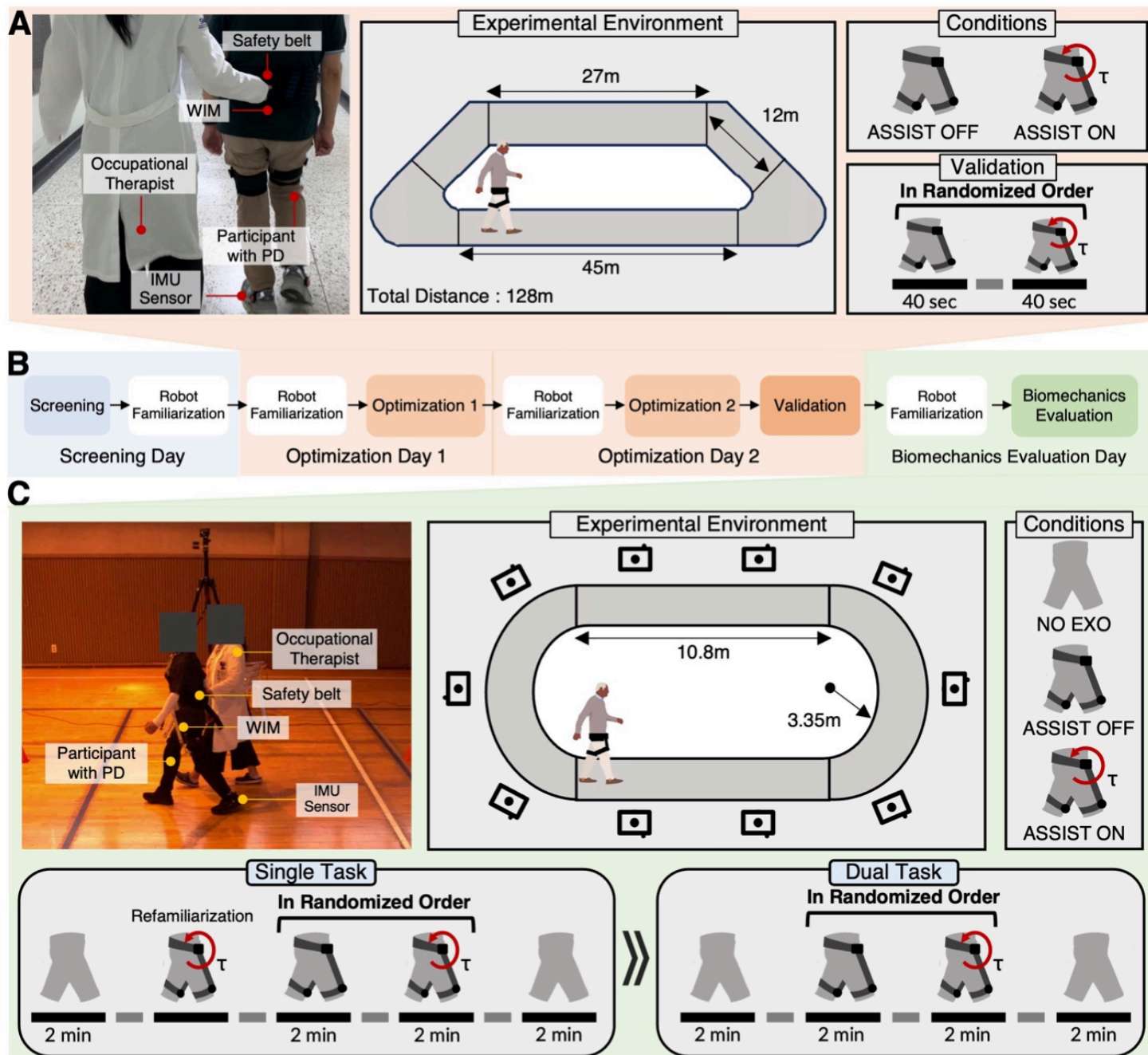


Figure 2

Overview of the study protocol and detailed procedures of Optimization Day and Biomechanics Evaluation Day. (A) Overview of the Optimization Day. The leftmost image shows the actual experimental setup during Optimization Day. (B) Overall experimental protocol of the study. (C) Overview of the Biomechanics Evaluation Day. The top-left image shows the actual experimental setup during Biomechanics Evaluation Day.

Optimized Control Parameters

Corresponding Assist Torque Outputs

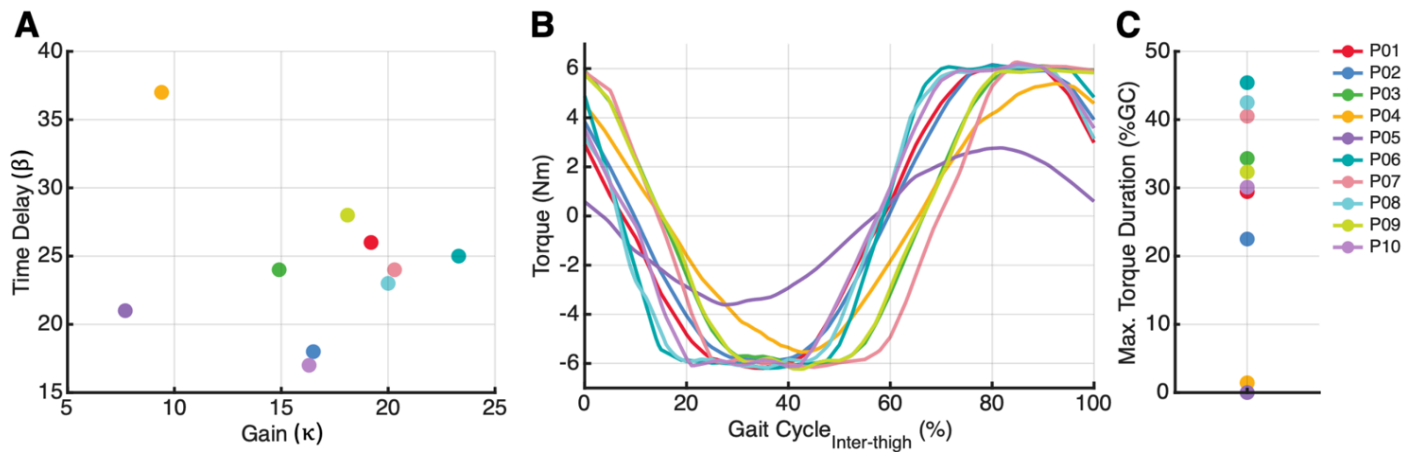


Figure 3

Optimized assistance characteristics across participants. (A) Optimized WIM control parameters—gain (κ) and time delay (β)—across participants. (B) Optimized assistance torque profiles across participants. Torque profiles were segmented at the local maxima of the inter-thigh angle and averaged across strides for each participant. The local maximum of the inter-thigh angle was defined as 0% of the gait cycle. The inter-thigh angle was defined as positive when the left leg is leading. Positive torque indicates hip flexion assistance to the left leg and hip extension assistance to the right leg, whereas negative torque indicates the opposite. Equal and opposite torque was applied to the contralateral hip. (C) Duration of the maximum allowable torque (6 Nm) in the optimized torque profile across participants. Duration was quantified as the percentage of the gait cycle during which a 6 Nm plateau was maintained (%GC).

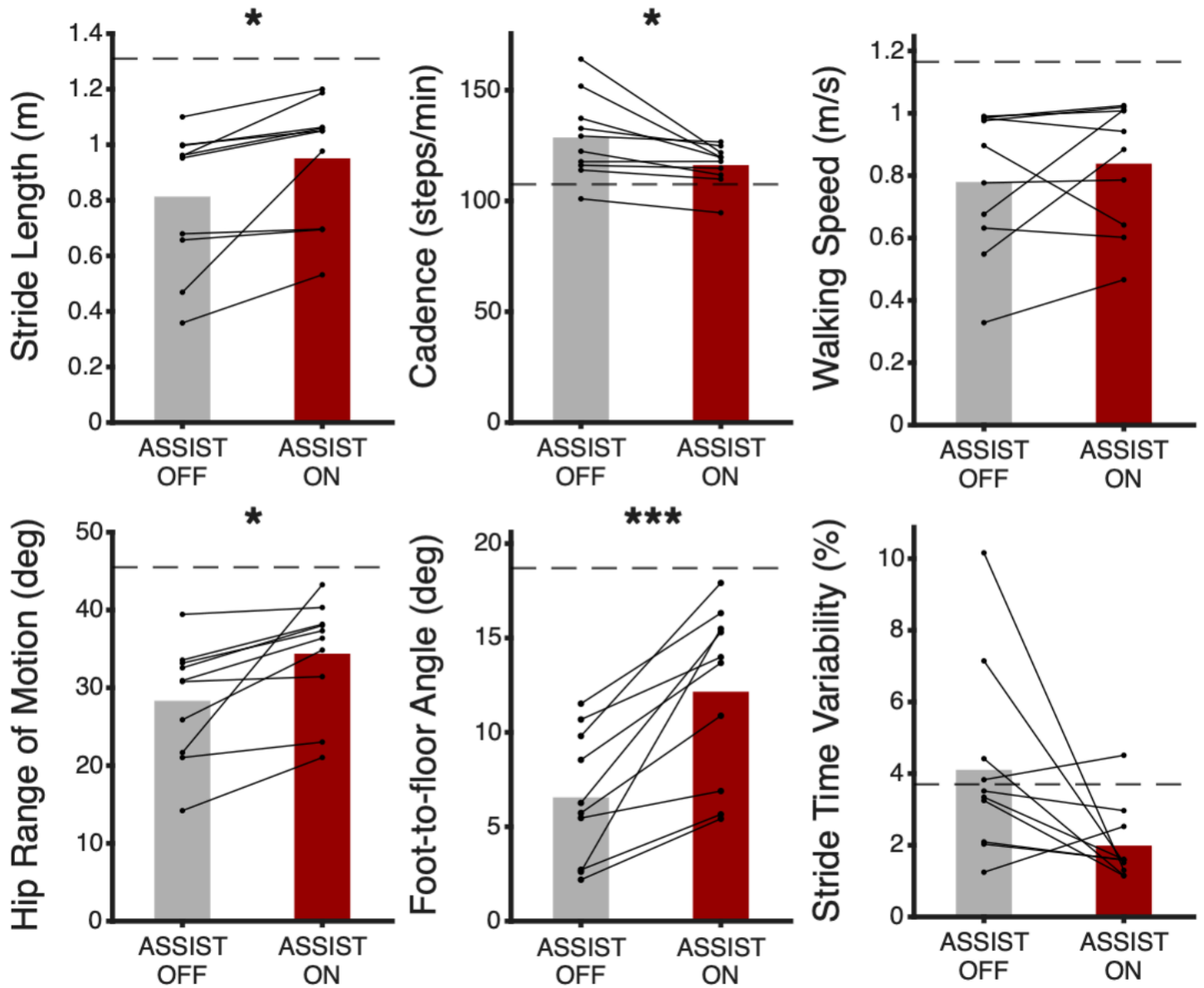


Figure 4

Changes in gait metrics during the Optimization Day. Bars represent group means, and black lines with markers indicate individual participants. The horizontal dashed line denotes the mean value of the healthy control group reported in the literature [42,43]. Individual participant values are provided in Supplementary Table 1. Significance markers: *** $p < 0.001$, * $p < 0.05$; only significant differences are indicated.

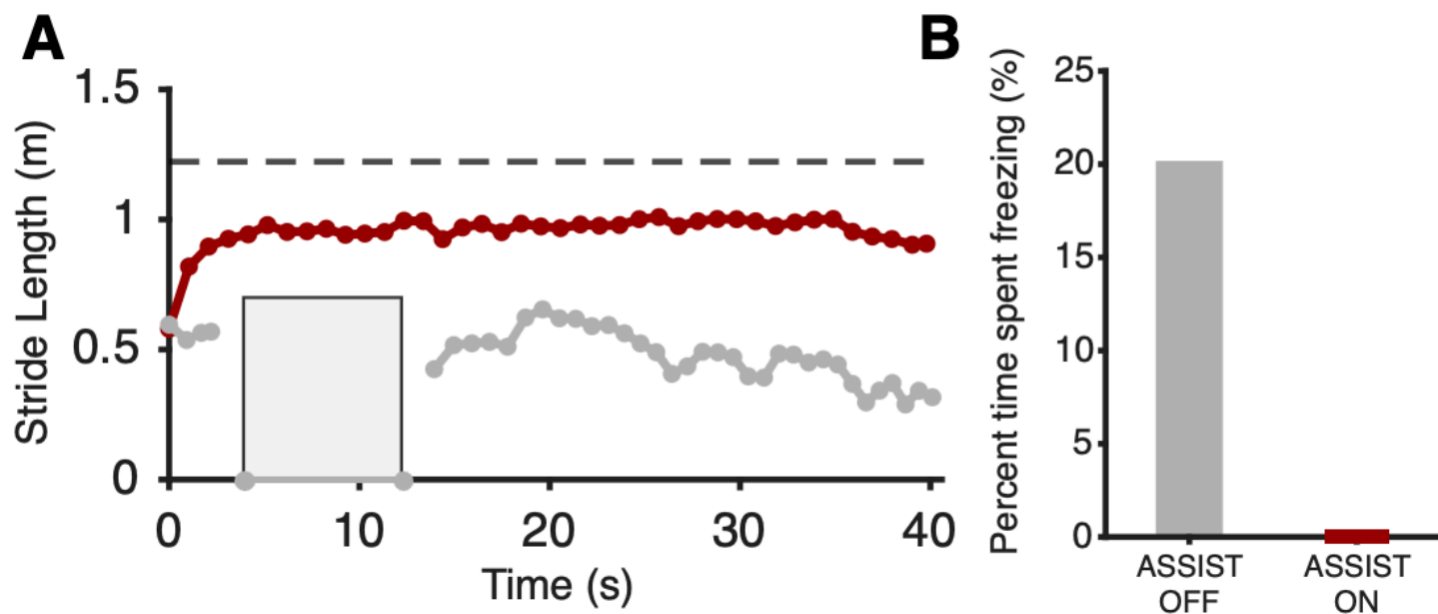


Figure 5

Comparison of freezing of gait (FoG) behavior under ASSIST ON and ASSIST OFF conditions in a participant with FoG (P05). (A) Time-series data of stride length during a 40-s walking trial. The horizontal dashed line indicates the age-, sex-, and height-matched stride length from healthy controls reported in the literature [41]. FoG intervals during the ASSIST OFF condition are highlighted with gray boxes. (B) Percentage of time spent freezing.

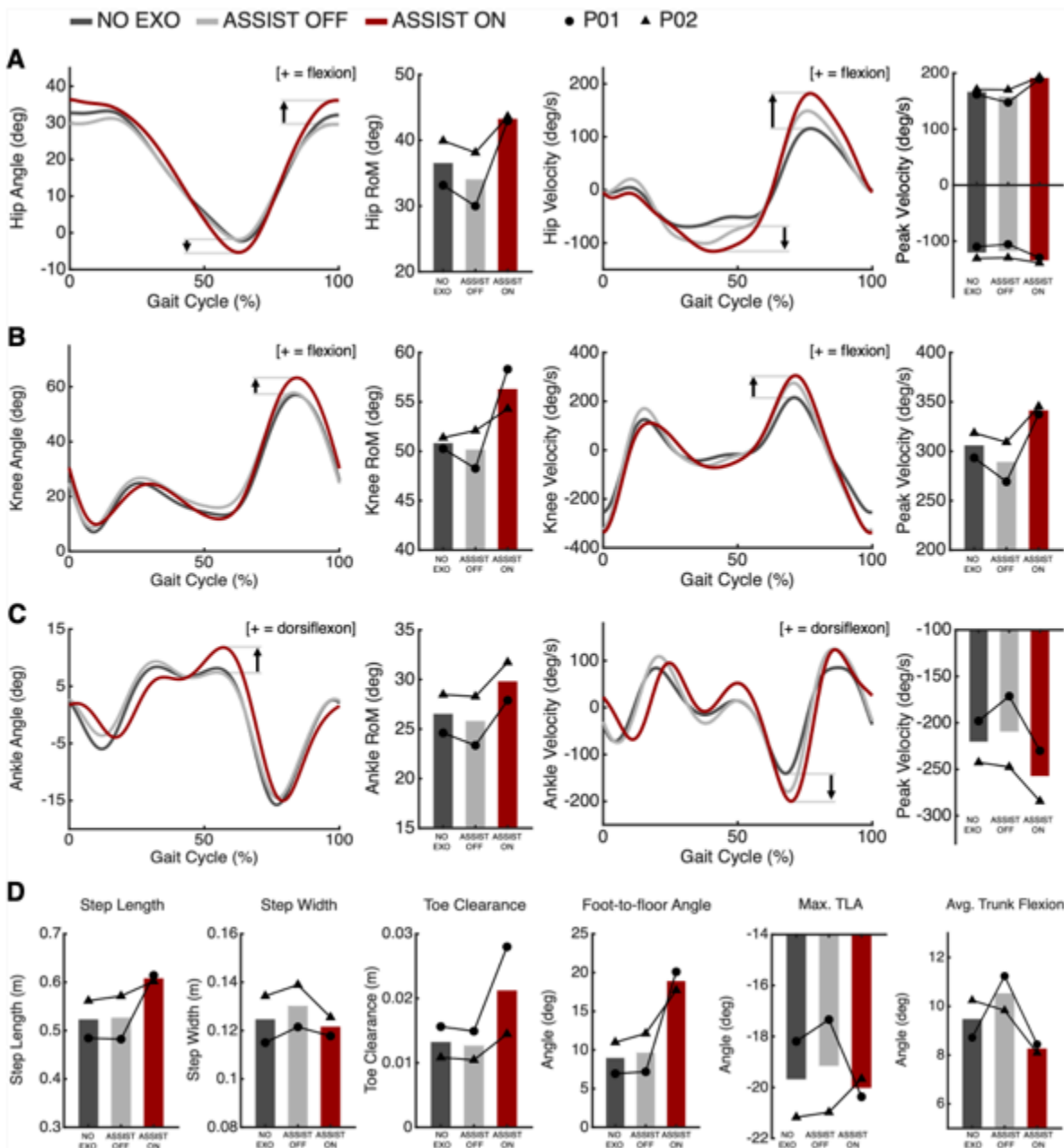


Figure 6

Effects of optimized assistance on lower-limb kinematics and trunk posture during the Biomechanics Evaluation Day. Panels A–C present lower-limb joint kinematics, including joint angle profiles, range of motion, velocity profiles, and peak velocities. Joint angle and velocity profiles were averaged across strides, sides, and participants. Profiles were segmented based on heel strike, with 0% of the gait cycle corresponding to heel strike. In each panel, the fourth plot shows peak velocities extracted from the velocity profile (third plot) at the peak values indicated by the horizontal reference lines, with arrows indicating changes associated with optimized assistance. (A) Hip joint kinematics. Velocity peaks correspond to the loading response and pre-swing phases. (B) Knee joint kinematics. Velocity peak corresponds to the swing phase. (C) Ankle joint kinematics. Velocity peak corresponds to the push-off

phase. (D) Additional gait parameters. From left to right: step length, step width, toe clearance, foot-to-floor angle, maximum trailing limb angle (Max. TLA), and average trunk flexion.

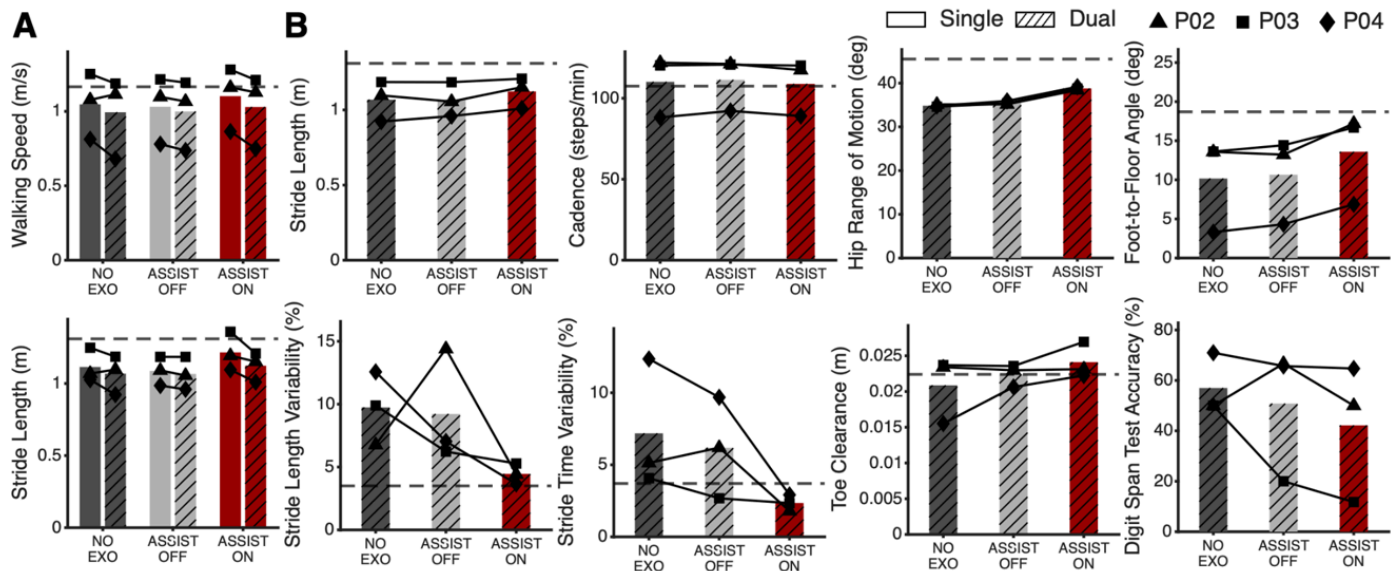


Figure 7

Changes in gait metrics and cognitive task performance under dual-task conditions with optimized assistance. (A) Comparison of gait metrics between single-task and dual-task conditions. (B) Effects of assistance on gait metrics and cognitive task performance during dual-task conditions. Group mean values are presented as bars, with individual subject data overlaid as markers. Single-task results are shown without hatching, whereas dual-task results are indicated with hatched bars. The horizontal dashed line denotes the mean value of the healthy control group reported in the literature [42–44].

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [SupplementaryMaterials.pdf](#)
- [SupplementaryVideo1.mp4](#)
- [SupplementaryVideo2.mp4](#)